

Comparative Study of Layer-Reduction Approaches

Sumit Jha

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Abstract

We implemented fifteen layer-reduction ideas for GPT-2 (124M) and SmolLM-135M and evaluated each on wikitext-2 subsets with a fixed PPL protocol. Four produced a usable signal: the fitted-ternary TerGRASP bridge (01), the ShortGPT BI baseline (06), SVD low-rank truncation (07), and the quantized no-op bypass (15). The rest failed by orders of magnitude (02, 08, 14), stayed at the baseline by construction (05, 11), or depended on a training budget we could not afford on one MPS laptop (03, 09, 10, 12, 13). This file walks through all fifteen, then gives the maths of each of the four working ones and the argument for taking 01 as the headline.

1 Setup used for every approach

Models: GPT-2 small and SmolLM-135M, loaded with `use_cache=False`. Data: Sales-force/wikitext, config wikitext-2-raw-v1 unless noted; train capped at 300k chars, validation/test at 100k chars. Evaluation:

$$\text{PPL} = \exp\left(\frac{1}{N} \sum_{i=1}^N -\log p_{\theta}(x_i \mid x_{<i})\right)$$

computed over non-overlapping 256-token windows, first 3000 tokens. Every number in this document is the PPL after the approach is applied to the lowest-BI layers (see Section 2).

2 BI ranking, common to most approaches

For decoder block l we record

$$\text{BI}_l = 1 - \mathbb{E}_t[\cos(h_t^{(l)}, h_t^{(l+1)})]$$

averaged over the batch and sequence. Low BI_l means the block is near the identity $h^{(l+1)} \approx h^{(l)}$, so it is the natural first candidate for deletion, merging, or aggressive quantization. GPT-2's five most redundant blocks are 4, 1, 2, 3, 5; SmolLM's are 2, 27, 5, 4, 3.

3 What we ran and how each came out

3.1 01 - Fitted-ternary TerGRASP (WORKS, headline)

Idea. For each redundant block L , collect $X = h_{in}, Y = h_{out}$ on $\sim 4\text{k}$ calibration chars, fit

$$\min_{U,V} \|XUV - \Delta\|_F^2, \quad \Delta = Y - X,$$

then AbsMean-ternarize U, V :

$$\gamma = \frac{1}{mn} \|W\|_1, \quad W_q = \gamma \text{clip}(\text{round}(W/\gamma), -1, 1), \quad W_q \in \{-\gamma, 0, \gamma\}.$$

The block is replaced by $g(x) = x + x U_q V_q$. **Why it works.** At a low-BI position Δ is small and lives on a low-dimensional manifold of directions, so a rank-32 factor recovers most of it; the ternary noise adds $\mathcal{O}(\gamma^2 \|x\|)$. **Result.** GPT-2 $k=1$: 45.11 vs 45.76 hard removal; SmolLM $k=1$: 21.71 vs 19.93. Tracks hard removal for $k \geq 2$.

3.2 06 - ShortGPT BI baseline (WORKS, the control)

Idea. Remove the k lowest-BI blocks; the control removes k random ones. **Result.** $k=1$ is free either way (GPT-2 45.76 vs 44.42 random); from $k=2$ on targeted removal dominates: GPT-2 $k=2$ 47.95 vs 139.09, $k=3$ 121.81 vs 4181.21. SmolLM $k=4$ 34.20 vs 57.37 random. **Role.** The falsification control: every other approach must beat or match its BI-targeted curve.

3.3 07 - SVD low-rank truncation (WORKS on SmolLM)

Idea. For each Linear W in the two most redundant blocks, $W \approx U_r S_r V_r^\top$ with rank r . **Maths.** By Eckart-Young the reconstruction error is the tail $\sum_{i>r} \sigma_i^2$ of the singular values; it is small when W is numerically low-rank, which empirically coincides with low-BI blocks. **Result.** SmolLM $r=256$: 19.93 (baseline), $r=128$: 22.00, $r=64$: 24.22. GPT-2 rows are a dtype-guard artifact and were archived for a clean fp32 rerun.

3.4 15 - Quantized no-op bypass (WORKS, supporting rule)

Idea. Quantize every weight inside the three lowest-BI blocks to $b \in \{8, 4, 2\}$ bits, AbsMean symmetric, and measure PPL. **Maths.** If $F_L(x) = x + \Delta(x)$ with $\|\Delta\|$ small, the layer’s output is insensitive to bounded weight perturbations $\|W - W_q\|_\infty \leq \gamma$: the induced change in output is $\leq \|\Delta W\| \|x\|$, tiny next to $\|x\|$. **Result.** SmolLM L5: 8-bit 17.76, 4-bit 17.77, 2-bit 17.76. SmolLM L2: 19.11/19.07/19.18. GPT-2 flat at 38.25 (eval noise; larger windows needed). **Rule.** BI answers both questions: "can I delete this?" and "can I crush its precision?".

4 What failed, with the exact failure mode

4.1 02 - Affine bridge across the deleted gap

We fit one ridge affine map $Y \approx AX + b$ from the input of the first removed block to the output of the last and inserted it as a single `nn.Linear`. **Why it fails.** The true map $F_{last} \circ \dots \circ F_{first}$ is not affine (it is a composition of LayerNorms, matmuls, and nonlinearities), so the single line cannot approximate it; PPL jumped to 4.5×10^3 (GPT-2 $k=1$) and 1.9×10^8 (SmolLM). **This is the direct motivation for 01.**

4.2 08 - Parameter soup of adjacent blocks

$\theta_{merge} = \alpha \theta_L + (1 - \alpha) \theta_{L+1}$. **Why it fails.** Weight-space averaging across a neuron-permutation boundary is meaningless; GPT-2 $k=1$ gave 103-252 PPL vs baseline 40.

4.3 14 - Cross-layer weight tying

Assigning `blocks[l+d] = blocks[l]` shares one set of weights between two positions. **Why it fails.** Depth is not redundant in weight space; tying at any period d exploded PPL to 10^3 - $10^6 \times$ baseline.

4.4 04 - Permutation-aligned averaging

Capture per-neuron activations of both adjacent blocks, run Hungarian matching to align neurons, then average. **Why it fell back.** The Hungarian step was not hardened up and the fallback was the naive average, so the result equals 08 and inherits its failure.

4.5 05, 10, 11, 12, 13 - scaffolds

Each of LayerDrop training (11), learned L0 gates (10), attn/FFN pruning proxy (12), early exit (13), and the 04-stage pipeline (05) executed but produced flat/identical rows because the key training loop (exit-head training, gate-sparsity thresholding, real LayerDrop fine-tune) was not run inside the free MPS budget. They are kept in **archived/** as incomplete.

4.6 03, 09 - distillation at a tiny budget

We wired KL-to-teacher (03) and KD-to-shallow-student (09) correctly but the 60-step AdamW pass on MPS made the KL loss essentially zero, so post-repair PPL equals pre-repair. The code path is reusable; only the budget was missing.

5 The working four compared head-to-head

Approach	PPL at operating point	Params effect	Training?	Note
01 Fitted-ternary TerGRASP	GPT-2 k=1 45.11, SmolLM 21.71	$\sim 20\times$ smaller on replaced block	yes, 80 steps	headline
06 BI baseline (targeted)	GPT-2 k=2 47.95, SmolLM 26.13	block removed	none	control
07 SVD r=128 (SmolLM)	22.00 vs 19.93 baseline	2-4 $\times$ on shrink blocks	none	smooth trade
15 2-bit on lowest-BI (SmolLM)	17.76 vs 16.64 baseline	$\sim 16\times$ on that block	none	rule

6 Why 01 is the headline

The other three working approaches each do one half of the job: 06 deletes (no compression of the kept graph beyond removal), 07 shrinks in width but keeps depth, 15 quantizes but keeps depth. 01 is the only one that simultaneously (a) keeps the graph depth, (b) shrinks the replaced block to ~ 1.58 bit/param, and (c) holds PPL at the baseline operating point $k=1$ on both models. Its failure mode is also the most defensible: at $k \geq 3$ it tracks hard removal instead of exploding, so the claim is bounded rather than brittle. That, plus the clean falsification control of 06 and the supporting rules of 07 and 15, is why we recommend 01.